

Sequential realization of Quantum Instruments

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In adaptive quantum circuits classical results of mid-circuit measurements determine the upcoming gates. This allows POVMs, quantum channels or more generally quantum instruments to be implemented sequentially, so that fewer qubits need to be used at each of the N measurement steps. In this paper, we mathematically describe these problems via adaptive sequence of instruments (ASI) and show how any instrument can be decomposed into it. Number of steps N and number of ancillary qubits n_A needed for actual implementation are crucial parameters of any such ASI. We show an achievable lower bound on the product $N.n_A$ and we determine in which situations this tradeoff is likely to be optimal. Contrary to common intuition we show that for quantum instruments which transform n to $m(> n)$ qubits, there exist N -step ASI implementing them just with $(m - n)$ ancillary qubits, which are remeasured $(N - 1)$ times and finally used as output qubits.

I. INTRODUCTION

Most of the currently developed quantum computers aim at implementing quantum circuits. In this paradigm a product pure state of n qubits is prepared and transformed by a sequence of unitary gates acting simultaneously only on a few qubits. Finally, projective single-qubit measurements are performed on all or just some qubits to signalize the outcome of the performed task. However, for many relevant tasks for quantum information processing, we start with a higher-level description (often less detailed) of what needs to be implemented. For example, preparation of a mixed state, realization of a quantum channel or a positive operator valued measure (POVM), or implementation of a quantum instrument (i.e., measurement having classical as well as quantum output). These concepts are mathematically more convenient when solving some optimization problems, since they focus only on the most general evolution of the primary quantum system, which might be effectively arising in many different ways from interactions with other classical or quantum systems. On the other hand, these concepts need to be realized on real quantum hardware via its available toolkit. For example, a quantum channel dilation is realized by appending ancillary qubits, and then performing a multi-qubit unitary gate, which is further decomposed into available quantum gates.

Current Noisy intermediate-scale quantum (NISQ) era devices [1] have serious limitations especially in terms of number of available qubits, number of gates or mid-circuit measurements that can be performed with reasonable quality within a single experimental run. Thus, to utilize potential of NISQ devices efficient realizations of channels, POVMs and instruments should be developed.

Although single-step dilations for POVMs [2], channels [3], and instruments [4] are known, their demands for a number of ancillary qubits often turn out to be too high [5, 6]. A practical strategy to solve this problem is to partially exchange space resources (qubits) for time

resources (the number of computational steps). One of the first efforts in this direction was made by Lloyd and Viola [7], who showed that repeated interaction with a single ancillary qubit, plus its measurement and reset, together suffice to sequentially realize an arbitrary quantum channel. This could be considered as one of the first examples of adaptive quantum circuits. Applicability of this approach was also considered for particular tasks, such as sequential implementation of POVMs [8, 9], tomography [10–18], computing [19, 20], state discrimination [21], joint measurability [22] and unambiguous state discrimination [23]. Design of such experiments for superconducting circuits, in particular by Chao Shen, et.al. [24], brought further progress including initial results for sequential implementation of quantum instruments.

Experiments of Córcoles, et.al. [25] on Quantum Phase estimation and P. Ivashkov, et. al. [26] on POVM implementation used adaptive quantum circuits, which performed comparably or outperformed the corresponding nonadaptive implementation. These two works clearly demonstrated that the current-era quantum technology could benefit from trading-off spatial to temporal resources, when designing the experiments. However, so far there was no systematic way of gradually exchanging the two resources.

The aim of the present work is to theoretically analyze adaptive quantum circuits in its idealized error-free form. Mid-circuit measurements are mathematically described by quantum instruments. Therefore, we begin by defining *adaptive sequence of instruments* (ASI), which we designed to be a sufficiently general structure to encompass the considered situation. We give it a new name due to its unique way of handling the classical feedforward of information. We explore the possibilities for the sequential implementation of quantum instruments by such sequences. In particular, we generalize such previous results [7, 9, 23, 24] and show a constructive decomposition technique for a quantum instrument into a N -step ASI based on sequence of POVM postprocessings of the instrument's induced POVM. We study ef-

efficiency of such sequential implementation schemes and determine achievable trade-offs by the proposed method between the number of steps N and the repeatedly used ancilla dimension needed for dilation of the involved instruments.

In the following we will concisely define needed concepts and the results we obtained. More detailed discussion and proofs can be found in the full manuscript [27].

Let $\mathcal{H}$ and $\mathcal{K}$ be two Hilbert spaces. General probabilistic transformations of quantum states, are termed *quantum operations*, and they are described by *Completely Positive Trace-Non-Increasing Maps* from $\mathcal{L}(\mathcal{H})$, denoting linear operators on $\mathcal{H}$ to $\mathcal{L}(\mathcal{K})$. *Quantum channel* is such quantum operation, which is deterministic in nature, and it corresponds to *Completely Positive Trace-Preserving Map* from $\mathcal{L}(\mathcal{H})$ to $\mathcal{L}(\mathcal{K})$. We denote the set of quantum channels as $Ch(\mathcal{H}, \mathcal{K})$. Each quantum operation $\mathcal{N} : \mathcal{L}(\mathcal{H}) \rightarrow \mathcal{L}(\mathcal{K})$ has a Kraus representation [28]: $\mathcal{N}(\rho) = \sum_m K_m \rho K_m^\dagger$, such that $K_m : \mathcal{H} \rightarrow \mathcal{K}$, $m = 1, 2, \dots, r_N$ and $\sum_m K_m^\dagger K_m \leq I$. Here, the smallest achievable number $r_N \in \mathbb{N}$ is called the Kraus rank of $\mathcal{N}$. If $\sum_m K_m^\dagger K_m = I$, the operation is a quantum channel. While quantum operations describe the mapping between initial and final state of the system they do not include any description of how such operation was actually implemented.

Quantum instrument describes a measurement procedure in such a way that it captures not only the measurement statistics, but it describes the post-measurement state as well. In order to achieve this, every possible outcome of an instrument is associated with a quantum operation. Thus, also quantum instruments do not include details of how measurement was performed, just its overall effect on states. For finite dimensional Hilbert spaces and finite outcome spaces quantum instrument can be defined [29–31], as a mapping $\mathcal{I} : i \mapsto \mathcal{I}_i$ from outcome set Ω to a set of quantum operations from $\mathcal{L}(\mathcal{H})$ to $\mathcal{L}(\mathcal{K})$ such that $\phi^\mathcal{I} := \sum_{i \in \Omega} \mathcal{I}_i \in Ch(\mathcal{H}, \mathcal{K})$ is a quantum channel. This condition indeed guarantees that the total probability of getting an outcome is one, i.e. $\sum_{i \in \Omega} Tr[\mathcal{I}_i(\rho)] = Tr[\phi^\mathcal{I}(\rho)] = 1$. We will denote quantum channel $\phi^\mathcal{I}$ as the *induced channel* of the quantum instrument $\mathcal{I}$. It is useful to define also the *induced POVM* $A^\mathcal{I} \in \mathcal{O}(\Omega, \mathcal{H})$ of an instrument $\mathcal{I}$ by equation $Tr[A_i^\mathcal{I} \rho] = Tr[\mathcal{I}_i(\rho)]$. We denote the set of instruments, which have outcome set Ω and map from $\mathcal{L}(\mathcal{H})$ to $\mathcal{L}(\mathcal{K})$, by $Ins(\Omega, \mathcal{H}, \mathcal{K})$.

II. ADAPTIVE SEQUENCE OF QUANTUM INSTRUMENTS (ASI)

Our aim is to define an N step *adaptive sequence of instruments* $\mathcal{Q}$ in which choice of each instrument in the

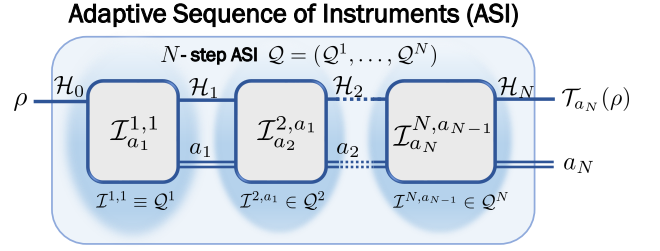


Figure 1: Illustration of N -step Adaptive sequence of instruments. Instrument in the k -th step is chosen based on the outcome a_{k-1} obtained in the previous step.

sequence depends directly on the classical outcome of the preceding instrument. At first sight it might seem restrictive to have dependence only on one preceding outcome. But, as we show in [27] actually arbitrary type of outcome dependence can be recast in this way via suitable choices of instruments and their outcome sets. Thus, we propose the following definition (see also Figure 1).

Definition 1. Let $\{\mathcal{H}_0, \dots, \mathcal{H}_N\}$ be an $(N+1)$ -tuple of Hilbert spaces, $\Omega_0 \equiv \{1\}$ and $\{\Omega_1, \dots, \Omega_N\}$ be a N -tuple of outcome spaces. We denote intermediate outcome at step k as a_k and we use upper index to signalize step number and to distinguish different instruments. Thus, let us choose sets of instruments $\mathcal{Q}^k \equiv \{\mathcal{I}_{a_{k-1}}^{k,a_{k-1}}\}_{a_{k-1} \in \Omega_{k-1}} \subset Ins(\Omega_k, \mathcal{H}_{k-1}, \mathcal{H}_k) \forall k, 1 \leq k \leq N$. The N -step adaptive sequence of instruments $\mathcal{Q}$ is an N -tuple of all sets $\mathcal{Q}^k$, i.e. $\mathcal{Q} \equiv (\mathcal{Q}^1, \dots, \mathcal{Q}^N)$.

Let us introduce the notion of Total Instrument by the following definition.

Definition 2. Let $\mathcal{Q}$ be N -step adaptive sequence of instruments. Then we define, a Total Instrument $\mathcal{T} \in Ins(\Omega_N, \mathcal{H}_0, \mathcal{H}_N)$ of $\mathcal{Q}$, as

$$\mathcal{T}_{a_N} = \sum_{a_{N-1} \in \Omega_{N-1}} \dots \sum_{a_1 \in \Omega_1} \left(\mathcal{I}_{a_N}^{N,a_{N-1}} \circ \dots \circ \mathcal{I}_{a_1}^{1,1} \right) \quad (1)$$

Total Instrument describes the overall instrument corresponding to the adaptive sequence of instruments $\mathcal{Q}$ if we forget (or ignore) the intermediate classical outcomes $a_1, \dots, a_{N-1}$. This approach is useful when we want the adaptive sequence of instruments to (deterministically) perform a fixed instrument with outcome set Ω_N . Thus, the Total Instrument $\mathcal{T}$ has the same outcome space as instruments in $\mathcal{Q}^N$.

In order to state our first result, we also need to recall the definition of POVM postprocessing.

Definition 3. POVM $B \in \mathcal{O}(\Omega_B, \mathcal{H})$ is called a postprocessing of POVM $A \in \mathcal{O}(\Omega_A, \mathcal{H})$, denoted by $A \rightarrow B$, if there exists a stochastic matrix $\nu_{kj} \ k \in \Omega_A, j \in \Omega_B$, i.e. $\nu_{kj} \geq 0, \forall k, \forall j$, satisfying $\sum_{j \in \Omega_B} \nu_{kj} = 1$ for all $k \in \Omega_A$, such that $B_j = \sum_{k \in \Omega_A} \nu_{kj} A_k \ \forall j \in \Omega_B$.

If the postprocessing matrix contains only values zero or one and the number of outcomes is reduced ($|\Omega_A| > |\Omega_B|$) we call such postprocessing a coarse-graining, since several outcomes are joined to a single outcome and the discrimination capability of the POVM decreases.

III. SEQUENTIAL REALIZATION OF QUANTUM INSTRUMENTS

Suppose we obtain instrument $\mathcal{T}$ as the composition of instruments $\mathcal{J}$ and $\mathcal{R}$. If we are asked to implement instrument $\mathcal{T}$, we can either use some dilation technique for it, or we could also consider creating a different dilation that can be built by combining dilations of instruments $\mathcal{J}$ and $\mathcal{R}$. This might be useful if we want to minimize the use of resources such as ancillary systems. In particular, if we are able to reset (reinitialize) the ancillary system, then the same qubits can be used for realizing the dilation of instrument $\mathcal{J}$ and for instrument $\mathcal{R}$. In this way, one can trade spatial resources (number of qubits) for temporal resources (number of steps to realize the task), since the dilations of $\mathcal{J}$ and $\mathcal{R}$ would be done one after another. Our goal in the rest of the manuscript is to explore how a fixed instrument $\mathcal{T}$ can be realized by an adaptive sequence of instruments. We will also refer to $\mathcal{T}$ as Total Instrument. The following theorem shows under which conditions we can always do that.

Theorem 1. *Given a Total Instrument $\mathcal{T} \in \text{Ins}(\Omega, \mathcal{H}, \mathcal{K})$ and any POVM $B \in \mathcal{O}(\Omega_B, \mathcal{H})$ such that the induced POVM of $\mathcal{T}$ can be postprocessed to POVM B (i.e. $A^\mathcal{T} \rightarrow B$), then there exists an adaptive sequence of an Initial Instrument $\mathcal{J} \in \text{Ins}(\Omega_B, \mathcal{H}, \mathcal{H})$ and a set of Residual Instruments $\mathcal{R}^j \in \text{Ins}(\Omega, \mathcal{H}, \mathcal{K}) \forall j \in \Omega_B$ such that*

$$\mathcal{T}_k = \sum_{j \in \Omega_B} \mathcal{R}_k^j \circ \mathcal{J}_j \quad \forall k \in \Omega \quad (2)$$

where the Initial Instrument $\mathcal{J}_j(\rho) = \sqrt{B_j} \rho \sqrt{B_j}, \forall \rho \in \mathcal{S}(\mathcal{H})$ is the Lüders instrument of POVM B .

Remark 1. *Let us note that outcomes j and k obtained in the sequential realization of the instrument $\mathcal{T}$ from Theorem 1 are not completely independent. One can prove that k can only take such values, for which the postprocessing matrix elements $\nu_{k,j} > 0$. In other words, the sequential realization is designed in such a way that some of the outcomes $k \in \Omega$ of the residual instrument $\mathcal{R}^j$ have zero probability of appearance for all states produced by quantum operation $\mathcal{J}_j$ of the Initial Instrument $\mathcal{J}$.*

Let us analyze the necessary ancilla size needed for sequential implementation of instruments proposed in The-

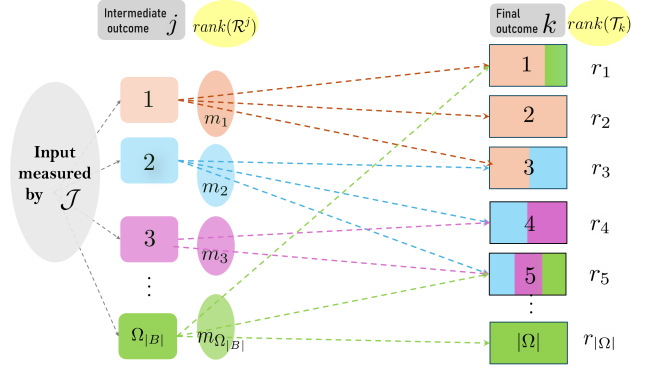


Figure 2: Illustration of how postprocessing matrix ν_{kj} with several nonzero entries in some of the rows k influences the Kraus rank of instruments $\mathcal{R}^j$ and consequently the dimension of the ancilla system d_A .

orem 1. Let's first concentrate on the case of non-shrinking Hilbert space ($\dim \mathcal{H} \leq \dim \mathcal{K}$). Let us denote by

$$g \equiv \lceil \dim \mathcal{K} / \dim \mathcal{H} \rceil \quad (3)$$

the minimal dimension of additional quantum system having Hilbert space $\mathcal{V}$ that needs to be added to allow embedding of Hilbert space $\mathcal{K}$ in $\mathcal{H} \otimes \mathcal{V}$. If $\dim \mathcal{K} = \dim \mathcal{H}$ then $g = 1$. If we assume the same ancillary system with dimension d_A can be used to realize instrument $\mathcal{J}$ and after reset any of the instruments $\mathcal{R}^j, j \in \Omega_B$ then it implies the following statements. The ancilla dimension d_A must be clearly at least the maximum of the needed dimension for the realization of the Initial instrument and any of the Residual instruments. For the instrument dilations we consider unitary transformation on the system plus ancilla, which is followed by projective measurement on the ancilla. As we rigorously show in the full manuscript [27] the subset of 2-step ASI originating from use of Theorem 1 and having minimal ancilla dimension always contains a setting in which $A^\mathcal{T} \rightarrow B$ is a coarse-graining, i.e. entries of ν_{kj} equal to zero or one. The difference in the needed ancilla size, which depends on the Kraus ranks of the involved quantum operations is illustrated on Figures 2 and 3. In case of coarse-grainings the needed ancilla size can be expressed as

$$d_A = g \max(\lceil \frac{|\Omega_B|}{g} \rceil, \{ \sum_{k \in \omega_j} r_k \}_{j \in \Omega_B}), \quad (4)$$

where r_k denotes Kraus rank of $\mathcal{T}_k$ and outcomes k of the Total instrument $\mathcal{T}$ are split into disjoint subsets $\omega_j \subset \Omega$ with $j \in \Omega_B$.

Optimization of d_A is in general quite complicated combinatorial task and it would be more pleasant to face the situation where each r_k is equal to one. Indeed, this can be done by the following trick. For each quantum instrument, $\mathcal{T}$, one can define many detailed quantum instruments [32]. These are instruments, which we obtain by

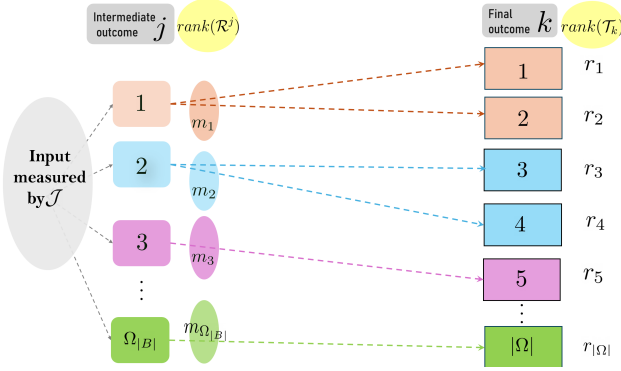


Figure 3: In contrast to Figure 2, if postprocessing matrix ν_{kj} contains only zeros or ones Kraus ranks of instruments $\mathcal{R}^j$ are generally smaller, since each r_k contributes to ranks of instruments $\mathcal{R}^j$ just once.

taking some Kraus representation for each quantum operation of instrument $\mathcal{T}$ and we define a single outcome of the new quantum instrument by each of all those Kraus operators. For the purposes of this paper it will be useful to consider the following subset of those instruments.

Definition 4. Suppose we choose some minimal Kraus representations $\mathcal{T}_k(\rho) = \sum_{m=1}^{r_k} T_{k,m} \rho T_{k,m}^\dagger$ for quantum operations $\mathcal{T}_k$ of quantum instrument $\mathcal{T} \in \text{Ins}(\Omega, \mathcal{H}, \mathcal{K})$. For every such choice we define set $\Omega_D = \{(k, m) | k \in \Omega, m = 1, \dots, r_k\}$ as the set of all tuples (k, m) and consequently we also define minimal detailed quantum instrument $\mathcal{T}^D \in \text{Ins}(\Omega_D, \mathcal{H}, \mathcal{K})$ by it's Kraus rank one quantum operations $\mathcal{T}_{k,m}(\rho) = T_{k,m} \rho T_{k,m}^\dagger$.

Obviously, by coarse-graining over the second element of 2-tuple outcome (k, m) of $\mathcal{T}^D$ we recover the Total instrument $\mathcal{T}$. Thus, if we would build an implementation of quantum instrument $\mathcal{T}^D$ on a real quantum hardware and out of obtained outcome (k, m) we would report just the value of k , addition of such simple classical postprocessing of outcomes would turn the experiment into an implementation of instrument $\mathcal{T}$. Performing an optimization of ancilla size given by Eq. (4) for detailed instrument leads to the following theorem.

Theorem 2. Consider quantum instrument $\mathcal{T} \in \text{Ins}(\Omega, \mathcal{H}, \mathcal{K})$ with Kraus rank $r_{\mathcal{T}}$ and such that $\dim \mathcal{H} \leq \dim \mathcal{K}$. Minimal ancilla dimension for implementation of quantum instrument $\mathcal{T}$ using 2-step ASI and classical postprocessing of outcomes is given by

$$d_A = g \lceil \sqrt{\frac{r_{\mathcal{T}}}{g}} \rceil, \quad (5)$$

where $g = \lceil \dim \mathcal{K} / \dim \mathcal{H} \rceil$ captures the ratio between dimension of output and input Hilbert space of the instrument $\mathcal{T}$.

Theorem 2 can be also interpreted as saying that for instruments with the same input and output Hilbert space

the graph of correlations between outcomes in successive steps of ASI (see Figure 3) is optimally an equally branching tree, which in effect minimizes the needed ancilla dimension.

IV. IMPLEMENTATION OF AN INSTRUMENT VIA N -STEP ASI

Using Theorem 1 recursively one can obtain the following result.

Corollary 2.1. Given a Total instrument $\mathcal{T} \in \text{Ins}(\Omega_N, \mathcal{H}_0, \mathcal{H}_N)$ and the postprocessing relations $A^{\mathcal{T}} \rightarrow B^{N-1} \rightarrow \dots \rightarrow B^1$ for POVMs $B^k \in \mathcal{O}(\Omega_k, \mathcal{H}_0)$ $k = 1, \dots, N-1$, there exists an adaptive sequence of instruments $\mathcal{Q} = (\mathcal{Q}^1, \dots, \mathcal{Q}^N)$ in which the instrument applied in the k^{th} step is denoted by $\mathcal{I}^{k, a_{k-1}} \in \mathcal{Q}^k$, the Initial instrument $\mathcal{I}^{1,1}$ is the Lüders instrument of POVM B^1 and

$$\mathcal{T}_{a_N} = \sum_{a_{N-1} \in \Omega_{N-1}} \dots \sum_{a_1 \in \Omega_1} \mathcal{I}_{a_N}^{N, a_{N-1}} \circ \dots \circ \mathcal{I}_{a_2}^{2, a_1} \circ \mathcal{I}_{a_1}^{1,1}. \quad (6)$$

The resource consideration leading to Theorem 2 can be generalized to the N -step scenario and provides the following exact formula

$$d_A = g \lceil \sqrt[N]{\frac{r_{\mathcal{T}}}{g^{N-1}}} \rceil = g \lceil \frac{1}{g} \sqrt[N]{g r_{\mathcal{T}}} \rceil. \quad (7)$$

For qubit systems it is useful to formulate the result in the following form.

Corollary 2.2. Given a Total instrument $\mathcal{T} \in \text{Ins}(\Omega_N, \mathcal{H}_0, \mathcal{H}_N)$ with $\dim \mathcal{H}_0 \leq \dim \mathcal{H}_N$ any N -step adaptive sequence of instruments, which implements $\mathcal{T}^D$ using Corollary 2.1, obeys the inequality

$$N n_A \geq \log_2 (g r_{\mathcal{T}}) \quad (8)$$

that requires the number of steps of the considered ASI times the number of used ancilla qubits n_A to be at least the logarithm of the product of the dimension increase factor $g = \lceil \dim \mathcal{H}_N / \dim \mathcal{H}_0 \rceil$ and the rank of Total instrument.

Previous considerations can be slightly generalized to characterize implementation possibilities of arbitrary N -step ASI, not just those produced via Corollary 2.1. It turns out (see [27]) that for instruments with $\dim \mathcal{H} \leq \dim \mathcal{K}$ the qualitative tradeoff $N n_A \geq \text{const}$ is unavoidable unless we can lower dimensions d_k significantly below d_0 , which does not seem to be feasible in general.

On the other hand, the dimension of the ancilla must be at least g (see Eq. (7)). One can observe that this lower bound can be reached for high enough N .

Corollary 2.3. *For any instrument $\mathcal{T} \in \text{Ins}(\Omega, \mathcal{H}, \mathcal{K})$ with $g = \lceil \dim \mathcal{K} / \dim \mathcal{H} \rceil > 1$ there exist N such that it can be implemented with N -step adaptive sequence of instruments (plus final postprocessing of classical outcomes), which does not require other ancillary systems than those needed to form the final output space ($d_A = g$). In particular, one can choose POVMs in Corollary 2.1 so that after the $(N - 1)$ -th step of ASI ancilla is reset and reused to realize an isometry conditionally chosen based on the obtained outcome a_{N-1} . Finally, outcome a_{N-1} needs to be coarse-grained back from Ω_D to the set Ω .*

V. SUMMARY AND DISCUSSION

Presented results are focused on optimization of space resources while implementing the desired quantum evolution. Our approach is based on describing mid-circuit measurements as quantum instruments, thus capturing both statistics of outcomes as well as the state change induced by such measurement. In particular, we have defined the notion of N -step adaptive sequence of instruments (ASI), which describes any quantum evolution interleaved with N measurements. Quantum instruments can have different dimension of input and output Hilbert space, or cardinality of the outcome space, which allows then to be equivalent to quantum states, channels or POVMs if these parameters are suitably chosen. Thus, the results we obtained are valid also for sequential realization of channels and POVMs. Central question is to find ASI for a fixed Total instrument that we intend to realize efficiently. Theorem 1 and its proof constructively show how this can be done starting from any postprocessing of the induced POVM of the Total instrument. In Corollary 2.1 we generalized this result to the N -step case and known schemes for sequential implementation of POVMs [9, 23, 26], channels or instruments [24] can be seen as its special case.

On the NISQ device one still needs to realize each quantum instrument in the ASI by using ancillary system, unitary transformation and a projective measurement on some subsystem. Here we assume quantum systems, can be reset and reused [33]. We analyze the dimensionality of the ancillary systems needed in all branches and steps of the evolution taking into account also the differences in the input/output dimension. We found that coarse-grainings, i.e. postprocessings which just join outcomes, in general lead to smaller ancilla dimension in applications of Theorem 1. For Total instruments, whose output Hilbert space dimension is not smaller than that of input, we determined the minimal ancilla dimension for the N -step ASI. Particularly relevant might be the following restatement of the result. ASI, which are constructed using Corollary 2.1 have due to Corollary 2.2 the number of mid-circuits measurements (steps of ASI) times

the number of used ancillary qubits lower bounded by a constant, which depends on the Kraus rank of the implemented instrument. These two Corollaries provide (up to our knowledge) the first practical way of trading off these two resources. Moreover, our considerations about general limitations of N -step ASI suggest that qualitatively this type of tradeoff is inevitable if the intermediate Hilbert space dimensions of the ASI can not be (sufficiently often) below the input dimension. Thus, the results of this manuscript could be seen as providing tools for performing tradeoff between number of qubits used (system plus ancilla) and the number of (intermediate) mid-circuit measurements.

Another interesting result is expressed by Corollary 2.3. In terms of qubits, we show that for any quantum instrument which transforms n to $m(> n)$ qubits, there exist N -step ASI implementing it just with $(m - n)$ ancillary qubits, which are remeasured $(N - 1)$ times and finally used as output qubits. In contrast, in previously published results on sequential decompositions of POVMs, channels or instruments increase of Hilbert space dimension was not considered and at least one qubit ancilla was needed in the studied cases.

Important parameter for NISQ devices is the number of used 2-qubit gates. Universal decompositions of state preparations, unitary transformations and isometries into quantum gates could be used for implementing instruments of the ASI. Unfortunately, these decompositions often yield exponentially long quantum circuits also for efficiently realizable transformations. This is the reason, why it is difficult to extend our analysis from the level of needed ancilla dimension to the number of needed gates. Conclusions based on universal decompositions may be true on most random instances of the problem (such approach was attempted by Iten, Colbeck, and Christandl in [34]), however in quantum computing we are typically interested in the set of measure zero, which applies only polynomial number of gates. Nevertheless, study of this workflow for medium system sizes and relevant quantum channels, POVMs or instruments might help to assess practical usefulness of the presented methods.

From a wider perspective (see Section 6 of the main manuscript [27]) the problem of finding 2-step ASI for a given Total instrument is related to compatibility of the Total instrument and a POVM. How this relation looks in the N -step case is an open question. Already for the 2-step case presented results do not need to provide exhaustive / optimal possibilities for construction of ASI, since (except for indecomposable instruments) they explore only a subset of compatibility possibilities. Thus, future research could explore potential of these uncovered cases. For example, ultimate trade-off bounds could be constructed if we would have better understanding of achievable Hilbert space dimensions in the ASI for a given Total instrument.

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